

Objective

How **prices move(ADR)** as the booking date approaches the check-in date

Table of Contents

- Assumptions
- Lay of the Land
- Deep-Dive(Messaging Strategy)
- Summary with Recommendations
- Next Step and Data Validation

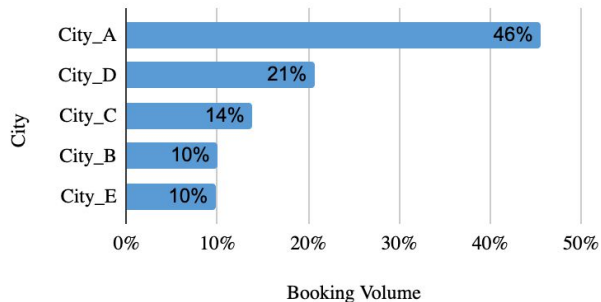
Assumptions

- **Unit of analysis:** Each line item represents a **unique booking (49K total bookings)**
- **Lead Time:** (Booking Date-Check in Date)
- **To measure price movement as check-in approaches, bookings are classified based on lead time (Check-in Date – Booking Date) into Early (11–60 days) and Last-minute (0–10 days), with price change computed as (Last-minute/Early)–1, to capture relative movement.**
 - a.) **Booking Classification:**
 - **Last-minute:** 0–10 days before check-in
 - **Early:** 11–60 days before check-in
- **Skewed ADR/Price Distribution** make global price thresholds unreliable; this is addressed by applying city-level benchmarking (median and 85th percentile) to define **Low, Budget-Friendly, and Luxury segments**
- **Ratings Segment:** High Rating(4.5+), Low Rating(<3) and Mid Rating(3-4.5) segments

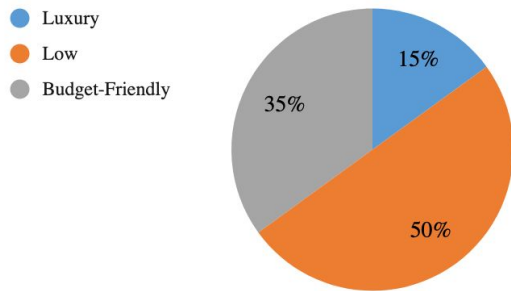
Lay of the Land

Are price movements mostly: discounting? stable? raising prices?

1. Booking Volume By City

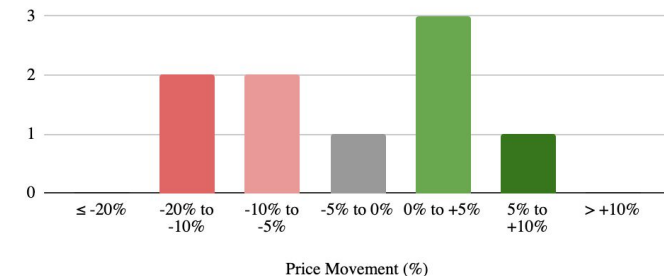


2. Segment Mix (By Price Segment)

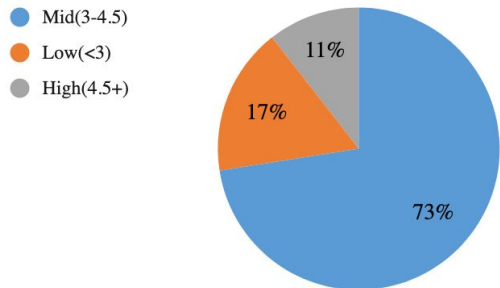


Current Price Movement Overview

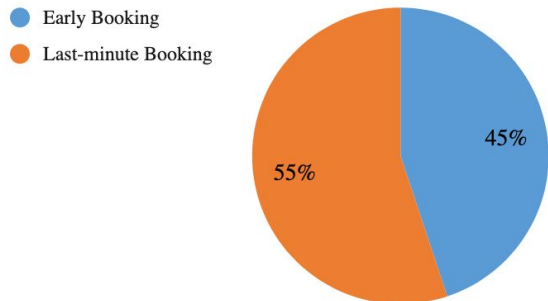
Discount Pressure(-ve) and ADR Strength(+ve) Study



3. Booking Volume By Rating Segment



4. Booking Volume By Type



1.Key Findings

48% of Demand is concentrated in City A and D
2 segments show high ADR Strength(+ve price movement)

2 segments show discounting pressure(-ve price movement)

2. Opportunity Areas: Opportunity lies in further City-level Segmentation(Segment and Ratings)

3. Next Step: Apply **Price Movement and Bookings Volume matrix by city segments** to determine high impact urgency areas

Price Movement vs. Booking Volume by City, Price & Rating Segment

High Volume + Price Decline(High Impact)

Discount/Deal Opportunity

Action: Use deal-focused messaging

City Segments	Price Change	Signal	Booking Volume
City A and C Low + Low Rating	City A: -18% City C: -10%	High ↑	4597, 9%
City B Low + Mid Rating	-10%	High ↑	2646, 5%
City A Budget + Mid Rating	-7%	High ↑	3819, 8%

City Segments	Price Change	Signal	Booking Volume
City D, E and B Low + Low Rating	City D: -13% City E: -15% City B: -31%	Low ↓	9000, 19%
City E Luxury + Mid Rating	-5%	Low ↓	315

Low Volume + Price Decline

Monitor (Low Impact)

High Volume + Price Increase

Urgency Opportunity

Use urgency/scarcity opportunity

City Segments	Price Change	Signal	Booking Volume
City D Luxury + Mid Rating	8%	High ↑	3046, 6%
City E Low + Mid Rating	5%	High ↑	1900, 4%

City Segments	Price Change	Signal	Booking Volume
City D Budget+Low Rating	10%	Low ↓	531
City D Low+Mid Rating	27%	Low ↓	561
City E Budget+Mid Rating	8%	Low ↓	873

Low Volume + Price Increase

Monitor (Low Impact)

Index

Low Segment

Luxury Segment

High Impact

Low Impact

-%

Price Change (Last-min Booking/Early Booking - 1)

%

Tableau Dashboard Link [here](#) (Real-time)

Summary

Context: Users often defer travel bookings while monitoring price fluctuations.
In order to solve this, Pricing movement study into a **bookings conversion opportunity** with targeted nudges during Oct-Nov-Dec (OND) 2016 seasonality

Priority	Booking CVR increased → Plausible reasons	Bookings (Abs.)	Bookings (in %)
P0	City A & C: Low Price + Low Rating Segment Price Movement A: - 18% and C: -10%	4597	3% (City A) 6% (City B)
P2	City A (Unique Pattern) Budget-Friendly + Mid Rating Price Movement: -7%	3819	8%
P1	City B and City E: Low + Mid Rating City B → Price Movement: -10%, 2.6K City E → Price Movement: 5%, 1900	2646 1900	5% (City B) 4% (City E)
P4	City D(Unique Pattern): Luxury + Mid Rating Price Movement: 8%, 3046	3046	6%
			32%

Key Insight: **Lack of targeted city segment nudges is hurting booking conversion.**

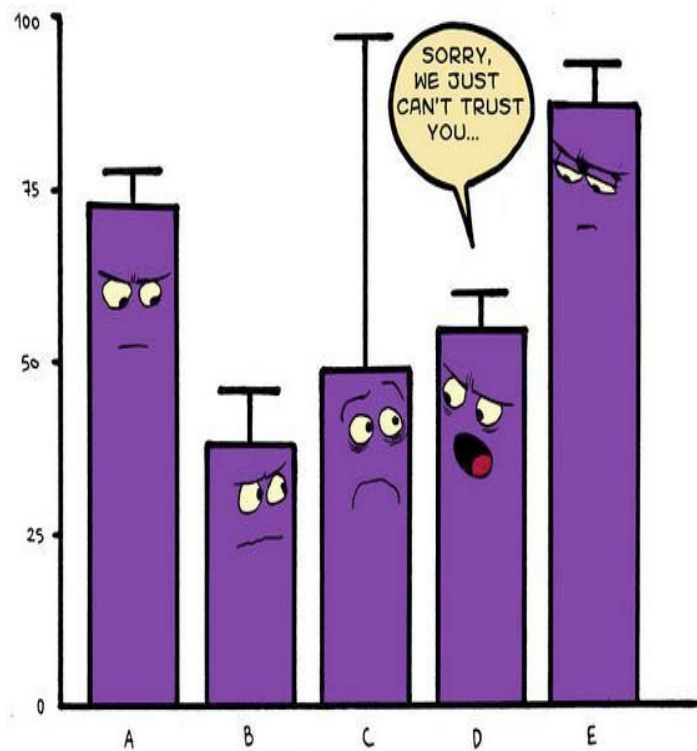
Recommendation/Opportunity Areas

S.No.	Booking CVR increased → Plausible reasons	Bookings (in %)	Price Behaviour Near Check-in	When to Trigger	Message Placement
1	City A (Low Price + Low and Budget + Mid Ratings)	11%	Low Price + Low: Decrease by -18% pp Budget + Mid:	Last Booking Useful(<10)	Checkout, Saved Bookings/Cart Page
2	City B (Low + Mid Rating)	5%	Decline by -10% pp	Last Booking Useful(<10)	Checkout, Saved Bookings/Cart Page
3	City C (Low + Low Rating)	6%	Decline by - 10% pp	Last Booking Useful(<10)	Checkout, Saved Bookings/Cart Page
4	City D(Luxury + Mid Rating)	6%	Increase by 8% pp	Early Booking Useful(<60)	Search Results Page, Booking Results Page
5	City E (Low + Mid Rating)	4%	Increase by 5% pp	Early Booking Useful(<60)	Search Results Page, Booking Results Page
6	Historical Data Analysis	20%	⚠️ Partial Urgency		
7	Remaining(Not stat. and practical sig.)	48%	Seasonality(data inconsistencies)	None	

100%

Note: Across ~42K bookings, any price movement ±5%(exclusive) is treated as seasonal variation; movements beyond this threshold are considered significant.

Next Step: Further Data Validation



Experimentation Design/Setup

- Run **A/B tests** to measure impact on booking conversion.
- Handle high variance (e.g., luxury segments) using **clustering/stratification**
- Explore variance reduction methods like **CUPED**

Success Metric:

- Increase in **booking conversion (incremental uplift)**
- **Primary: City segmentation to view where it's effective**

Guardrail:

- **Cancellations should remain stable**
- **Notifications fatigue should remain stable**

Statistical Setup:

- Measure **practical impact (MDE)** using historical benchmarks
- Use **95% confidence level** and **80% power**
- Allow:
 - **5% false positives** (detecting impact when there isn't one)
 - **20% false negatives** (missing a real impact)

Thankyou